Venkata Lakshmi Parimala Pasupuleti

Senior Backend Engineer · Distributed Systems · Regulated Financial Workflows · Python · AWS · Docker · Kubernetes · AI-Assisted Development

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PROFESSIONAL SUMMARY

Backend software engineer with 8 years owning product and platform development end-to-end in a **regulated financial data domain**: HUD/LIHTC compliance, income eligibility, HAP payment workflows, and rent schedule processing for 100K+ housing units - the same regulated workflow patterns that govern **mortgage servicing and mortgage transactions** at scale. At Entrata, designed and maintained **distributed infrastructure** including RabbitMQ async processing with idempotency guarantees, microservices architecture, AWS S3/RDS data pipelines, and Docker containerization - achieving 50% scalability improvement and 99.5% uptime under **regulatory requirements**. Expert in Python and SQL (200+ production query optimizations); shipped 20+ full-stack modules across 120+ Agile sprints in a **high-growth environment**. Daily user of **AI coding tools** (Claude Code, Cursor, GitHub Copilot) with rigorous human review gates. Built LeaseI (compliance audit automation platform, LangGraph + Python) and AgentProbe (AI evaluation toolkit with CI/CD quality gates). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Strong **product judgment**: consistently balanced engineering standards with delivery velocity.

TECHNICAL SKILLS

Backend & Languages: Python (expert), PHP, JavaScript, TypeScript, Java, R, SQL (expert — 200+ production optimizations), PL/SQL, Bash; FastAPI, Django, Spring Boot, Laravel, REST APIs, GraphQL

Distributed Systems & Cloud: AWS (S3, EC2, RDS, Glue, Lambda), Google Cloud Platform / GCP & Vertex AI (familiar), Docker, Kubernetes (K8s), CI/CD (GitHub Actions), RabbitMQ, Redis, Celery, Apache Kafka, Terraform

Data & Databases: PostgreSQL, MySQL, SQLite, Snowflake, Oracle, NoSQL, ETL/ELT Pipelines, Schema Design, Data Normalization, Data Governance, PHI / HIPAA Compliance, data structures and algorithms

AI Coding Tools & Dev: Claude Code (daily), Cursor, GitHub Copilot, LangGraph, LangChain, FastAPI, React.js

Architecture & Engineering Standards: Microservices, Multi-Tenant Architecture, System Design, SOLID/DRY, OOP, Dependency Injection, ORM Frameworks, Agile/Scrum (120+ sprints), Code Reviews, large-scale systems design

EXPERIENCE

Entrata (PropTech / Real Estate SaaS)

Sep 2018 - Aug 2024 · Remote

Senior Software Engineer - Affordable Housing Backend Platform

- **Owned backend product and platform development from design to production** - HUD compliance pipeline, income eligibility, HAP payment workflows, rent schedule processing, and tax-credit calculations for 100K+ housing units at 99.5% uptime. Built and maintained 20+ full-stack modules (Python, PHP, PostgreSQL, JavaScript) across 120+ Agile sprints in a high-growth PropTech environment.
- **Designed and owned distributed infrastructure for large-scale regulated workflows:** RabbitMQ async processing system for gross-rent updates across thousands of tenant records - designed retry logic and idempotency guarantees eliminating double-processing failures; owned through production incidents and annual rent-schedule cycles.
- **HUD income-limit import pipeline (owned 7 years):** ~3,000 federal source rows expanded into ~180,000 domain records powering eligibility, rent, HAP, and tax-credit calculations. Every annual release included baseline validation, reconciliation, and rollback preparation - zero compliance violations across the full ownership period. This is exactly the scale and rigor required for mortgage transaction processing under regulatory requirements.
- **Led AWS S3 document migration** for the Affordable Housing module - inventoried document types, mapped legacy records to migration framework contracts, validated content and metadata, confirmed customer-facing workflows post-cutover; 100% PHI compliance with Libsodium-encrypted pipelines; zero security incidents. Resolved 90% of legacy storage constraints.
- **Optimized 200+ SQL queries (30% performance gain)** through execution plan analysis, indexing, partitioning, and stored procedure refactoring across complex multi-tenant schemas for eligibility, financial, and document workflows - correctness-first approach: a fast query returning wrong financial data is a production failure.
- Led microservices redesign using Docker - **50% scalability improvement, 30% deployment time reduction**; Python/Bash automation eliminated 80% of manual operational workload. Mentored 5+ engineers; conducted 300+ peer code reviews with strict engineering standards.
- Daily user of **AI coding tools (Claude Code, Cursor, GitHub Copilot)** with disciplined human review gates - applied to architecture design, code quality, and system design decisions.

Entrata India (Db Xento Systems), Pune

Jan 2017 - Sep 2018

Associate Software Engineer / Trainee

- Built property management and compliance backend modules (PHP, Python, Java, PostgreSQL) - income limits import (1.6M HUD/Tax Credit records across 5 years), document management, and regulated compliance workflows for affordable housing.
- Led cloud migration of client documents to AWS S3; redesigned legacy document storage database architecture; 56 state-specific housing certification documents with merge fields; 6 months post-deployment stabilization.

Blue Horse Digital

Sep 2026 - Present · Virginia, USA

AI/ML Engineer (Volunteer)

- Developing ML models and backend data pipelines for business intelligence - feature engineering, model training, and backend API integration; applying AI coding tools with rigorous engineering standards throughout.

George Mason University

Aug 2024 - May 2026 · Part-time

Research Associate & Graduate Teaching Assistant

- Built multi-source prediction models (Python, R) on WHO and UK Road Safety datasets - $R^2 = 0.977$ regression accuracy; applied Isolation Forest for automated data integrity checks; fine-tuned a custom classification model (HuggingFace + PyTorch, deployed on Quart).
- Delivered Python and database workshops to 600+ students; GCP / Google Cloud Platform coursework (Vertex AI, GCS, BigQuery) - familiar with GCP infrastructure and deployment patterns.

PROJECTS

Leasel: Compliance Audit Automation Platform REGULATED DOMAIN PYTHON · LANGGRAPH

2026 | Python, LangGraph (8-agent), FastAPI, TF-IDF, OCR, SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production-grade compliance audit automation platform for regulated financial/housing workflows - directly maps to mortgage servicing audit and regulatory requirements. Core architecture rule enforced in code: LLM reasons; deterministic logic or human makes every final regulated call.
- Three-surface platform: Document Intelligence (OCR, section-aware chunking, hybrid TF-IDF + semantic search, citation grounding, human-review gate on regulated decisions); 8-agent LangGraph investigation pipeline with durable state and conditional HITL escalation; Tenant Self-Service (eligibility always routes to staff - never AI-resolved).
- Results on 1,240 residential lease PDFs across 5 US states + federal: **93.6% precision, 87.9% F1** across 10 compliance rules; pass@1 81.7%, pass@3 88.3% via regression harness with LLM-as-judge scoring.

AgentProbe: AI-Assisted Development Toolkit with Quality Gates ENGINEERING STANDARDS CI/CD · EVAL

2025 | Python, FastAPI, LangGraph, ChromaDB, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- Open-source AI evaluation toolkit designed to enforce engineering standards on AI coding tool outputs - 15-endpoint FastAPI backend, 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression harness with golden datasets, multi-provider LLM routing (OpenAI/Anthropic/Groq/Ollama). Runs 100% locally, zero vendor dependency.
- Built to apply rigorous human review gates on top of AI coding tool (Cursor, Claude Code, Copilot) outputs - the workflow I apply daily for production backend software engineering with high engineering standards.

CrashClarity: Large-Scale Analytics Engineering (104K Records) DATA · PYTHON

2025 | Python, R, scikit-learn, Random Forest, Logistic Regression, GitHub Pages | Portfolio

- End-to-end analytics engineering on 104K+ UK accident records - feature engineering across weather, road, and time variables; Random Forest + Logistic Regression pipeline (85%+ accuracy); interactive dashboard deployed on GitHub Pages. Demonstrates data structures and algorithms + large-scale data analysis proficiency.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 | GPA 3.87

Coursework: Distributed Systems, System Design, Data Structures & Algorithms, Cloud Architecture (GCP/AWS), Machine Learning, Database Engineering, Statistical Modeling.

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013-2017